PORTFOLIO RESEARCH / REFRESH & RE-UNDERWRITE / 19 OCTOBER 2026

Taiwan Semiconductor Manufacturing Co.

TSM · NYSE • ADR · ADVANCED FOUNDRY • AI INFRASTRUCTURE

ADD SELECTIVELY · RESEARCH CONVICTION 7/10

The moat is widening with N2 and AI advanced packaging, but the stock return now depends on translating extraordinary fab spending into sustained distributable cash flow.

Investment horizon: 3–5 years · Price: October 19 regular-session close · Preliminary scenario estimates, not a guaranteed 12-month target

MARKET CLOSE	BASE DCF	SCENARIO WEIGHTED	PORTFOLIO WEIGHT
$420.22	**$492**	**$548**	**0.98%**
19 Oct 2026, 4pm ET	+17% estimated vs. close	+30% vs. close	41 shares / ADRs

SOURCE-CONTROL SUMMARY

Quarterly financials through Q3 2026 · Oct 15; market data October 19; estimates sampled October 19; holdings, quantity, cost basis and weight solely from **retirement-portfolio-63.xlsx** uploaded by the user. Default-branch GitHub manifest still incorrectly names export 61; not used for portfolio numbers.

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01 · Executive investment judgment

Strong post-earnings operating evidence supports selective additions, not an assumption that every AI capacity dollar earns peak returns.

BUSINESS ECONOMICS

- Critical supplier to NVIDIA, Apple, AMD and hyperscaler silicon designs; multi-node tape-out ecosystem and yield learning are hard to replicate.
- N2 ramp and CoWoS/advanced packaging produce high-value wafer and packaging mix that should support long-duration operating margins if capacity stays scarce.
- Price increases, design scale and disciplined utilization can offset some geographic expansion costs.

PRINCIPAL BEAR COUNTERARGUMENT

- Overseas fab economics can dilute consolidated gross margins and lock in years of capex before acceptable returns are observable.
- AI customer concentration and cloud-provider internal silicon decisions can create correlated order revisions; N2 capacity is planned into an uncertain 2028 demand curve.
- Taiwan geopolitical constraints, trade controls and cross-strait transport risk are not captured by a normal equity discount rate.

Decision hinge. Current price **$420.22** compared with conservative FCFE DCF base **$492** (+17%) and 25/50/25 weighted scenario **$548**. These are 10-year present values, not near-term price promises. Proposed Equity Research posture **ADD SELECTIVELY**, conviction 7/10. CIO owns authoritative thesis status, capital allocation and trade decision.

COMPANY QUALITY

9.5 / 10

Moat, repeatability, margins, balance-sheet resilience, management and investor control quality.

VALUATION / PORTFOLIO FIT

7 / 10 / 7 / 10

Value versus expectations / incremental fit at current workbook concentration; distinct from business quality.

Scoring and recommendation are research assessments, not externally sourced credit ratings or a change to Portfolio CIO-controlled status.

02 · What has changed since previous underwriting?

Previous investment research: **October 15 post-Q3** (market anchor **$416.42**; prior base approximately **$540**). The company-specific developments below determine which model and thesis assumptions require refreshing.

Q3 outperformance	Revenue $46.72B exceeded the prior $44.6–45.8B outlook; operating margin 58.7% exceeded its previous guidance ceiling.
Advanced-node momentum	2nm reached 8% of wafer revenue, 3nm 33% and 5nm 28%; sub-7nm advanced processes reached 79% of wafer revenue.
Forward capacity economics	Q4 sales guidance is $52.3–54.3B; FY2026 capex $60–64B with meaningful additional overseas expansion and execution risk.
Valuation and entry reference	At $420.22, the shares are broadly flat against the prior ~$416.42 premarket price; prior $540 base was EPS-multiple-led rather than explicit owner-cash-flow DCF.

Price-anchor changes are period-specific and do not prove a thesis revision on their own. New DCF outputs should not be compared mechanically with earlier forward-P/E targets without noting methodology differences.

03 · Business quality, industry structure & moat

Core underwriting pillars

- Critical supplier to NVIDIA, Apple, AMD and hyperscaler silicon designs; multi-node tape-out ecosystem and yield learning are hard to replicate.
- N2 ramp and CoWoS/advanced packaging produce high-value wafer and packaging mix that should support long-duration operating margins if capacity stays scarce.
- Price increases, design scale and disciplined utilization can offset some geographic expansion costs.
- Net cash provides flexibility through a large capital cycle, but not all cash can be treated as available for shareholder distribution.

Strongest counterevidence and competitive pressure

- Overseas fab economics can dilute consolidated gross margins and lock in years of capex before acceptable returns are observable.
- AI customer concentration and cloud-provider internal silicon decisions can create correlated order revisions; N2 capacity is planned into an uncertain 2028 demand curve.
- Taiwan geopolitical constraints, trade controls and cross-strait transport risk are not captured by a normal equity discount rate.
- At ~$420, a large amount of future operating success is already recognized despite substantial near-term capex.

> A strong business can be a poor security when the entry price demands an unrealistically good path—or a high-return idea can fail because operating economics change. The contrary cases above are treated as alternative hypotheses to test, not as fine-print risks.

04 · Earnings, cash generation & management execution

Reported evidence and economically relevant operating metrics

Q3 2026 sales	$46.72B	Company Q3 SEC exhibit; +41.2% YoY in USD; +50.9% in NT$
Q3 net income	NT$763.06B	+68.7% YoY; US$4.60 earnings per ADR
Q3 gross / operating margin	66.4% / 58.7%	Versus Q2 67.7% / 60.3%
Advanced wafer mix (2nm/3nm/5nm)	8% / 33% / 28%	Management shipments by process generation
Q4 2026 sales guide	$52.3–54.3B	65–67% gross / 56.5–58.5% operating margins
Trailing operating cash / capex	$96.26B / $50.69B	Market-data LTM, approximate US$
Trailing reported FCF	$45.57B	CFO less capex; high capex sensitivity

LATEST KEY OPERATING FACTS

Q3 REVENUE
$46.72B
+41.2% YoY in US dollars

Q3 GROSS MARGIN
66.4%
Down 1.3pt sequentially

FY26 CAPEX
$60–64B
Management spending plan

2027 REV. CONSENSUS
~NT$7.65T
Taiwan-dollar estimate; FX-sensitive

GAAP quality, capital allocation, cash-flow bridge

Unlike high-margin pure software, the bulk of manufacturing gross profit is not distributable without recurring R&D, fab depreciation replacement and fresh wafer/packaging investment. Higher N2 yields and overseas-plant utilization are necessary to realize the cash growth path.

The report uses trailing FCF as a checked market-data starting point, but the DCF's economically sustainable year-zero FCF is explicitly identified as an *analyst scenario input*. Public consensus and company guidance are never substituted for an audited owner-earnings forecast.

05 · Earnings expectations, market valuation & peers

Adjusted ADR EPS, LTM	~$15.29	~27.5× current P/E
FY2027 consensus EPS	NT$148.61 / ordinary share	≈$23.36 per ADR at NT$31.8/$; FX assumption
2027 consensus P/E	~18.0×	Estimate and FX dependent
FY2027 estimated revenue	NT$7.65T	~37.5% growth; mix with SEC USD reporting carefully
Reported trailing P/FCF	~47.8× reconciled	Current close × 5.186B ADR units / $45.57B FCF
Estimated peer ranges	Intel / Samsung / leading-edge equipment	Non-like-for-like margins/capex; no simplistic single peer multiple
Reliable PEG	~0.52× illustrative	18× P/E / 34.9% forecast EPS growth; FX & cyclical revisions weaken precision

Comparison is drawn against history, investor consensus and economics of realistic alternatives, not a peer-multiple average: high-margin software, insurance services, leading-edge wafer manufacturing and two-sided marketplaces warrant different growth, maintenance capital and valuation measures. Any forward P/E cited uses provider-supplied or clearly labeled adjusted EPS, while GAAP EPS may differ materially. PEG is useful only when estimated earnings growth is consistent, sustainable and positive; it is not an overriding buy rule.

Market price versus expected 2027 performance

At **$420.22**, the equity market is judging durable future growth and margin, not merely whether the latest quarterly print beat consensus. Each valuation ratio has a timing, share-count and accounting-basis dependency. The next release date is **Q4 2026 release / monthly revenue**.

06 · Discounted cash flow, reverse DCF & cost-of-capital sensitivity

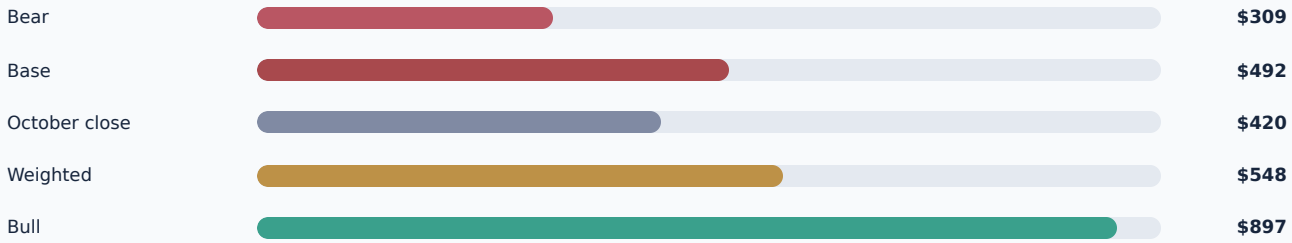
Ten-year equity free-cash-flow DCF — company-specific economics, explicit growth fade and cash ownership

> **Model architecture.** Start with normalized 2026 **$50.00B** of equity FCF; shares / ADR-equivalents **5.1860B**; equity discount rate **10.5%**; terminal growth **3.0%**; explicitly modeled excess cash **$70.0B**. Because the cash flow is after interest, net debt is not deducted again. Terminal value = year-10 FCFE × (1 + g) / (r − g), discounted ten years.

CASE	EQUITY DISCOUNT	YR 1 FCFE	YR 10 FCFE	DCF PER SHARE	VS. PRICE
Bear	11.5%	$62.5B	$204.7B	**$309**	-26%
Base	10.5%	$70.0B	$291.9B	**$492**	+17%
Bull	9.5%	$77.5B	$482.9B	**$897**	+114%

All cash flow values USD billions and intrinsic per-share amounts in USD. Case growth schedules and analyst discount-rate assumptions are disclosed below. The bull/bear change the discount rate ±100 basis points as well as growth; compare as joint scenarios, not isolated WACC estimates.

VALUE DISTRIBUTION



Scenario mix: bear 25%, base 50%, bull 25%. Weighted value $548; modeled base $492. Probabilities reflect research judgment rather than statistical estimates.

Reverse DCF: what is priced in?

A constant annual equity-FCF growth rate of **18.4%** for ten years would match $420.22 with the base discount rate and terminal assumption. Year-10 normalized equity FCF would be **$269.8B**. This is a market-implied growth hurdle under a deliberately simplified flat-growth trajectory.

Owner-earnings sensitivity

If starting distributable FCF is **15% lower** ($42.50B), the reverse growth hurdle becomes **20.6%** per year. This sensitivity addresses continuing reinvestment, capital allocation and share dilution but is not a second reported earnings figure.

Base intrinsic value sensitivity to cost of equity

DISCOUNT	EQUITY VALUE/SHARE	VERSUS CLOSE
9.5%	$578	+37.5%
10.0%	$532	+26.5%
10.5%	$492	+17.0%
11.0%	$457	+8.8%
11.5%	$427	+1.5%

Cash-flow provenance and normalization: Beginning FY2026 normalized equity cash flow is analyst-estimated $50B versus $45.57B TTM reported FCF; explicit $70B of excess liquidity is added (a conservative subset of the ~$100B market-data net-cash figure), and ordinary-share count is divided by five to obtain 5.186B ADR equivalents. 2027 capex can make this normalization too optimistic.

Model assumes no terminal disruption, constant terminal economics, stable projected share count and returns proportional to FCFE. If buybacks materially reduce diluted shares, FCFE per-share could improve; if dilution, acquisitions or insurance/fab funding absorbs more capital, intrinsic value may fall. A risk-free rate near 5% materially affects long-duration valuations.

07 · Bull / base / bear operating cases

BEAR • 25% PROBABILITY

$309 per share

Downside from slower adoption, margin/FCFE growth and higher required return.

First-year FCF growth: +25%; terminal-year growth: +6%; equity discount 11.5%.

BASE • 50% PROBABILITY

$492 per share

Durable execution with a deliberately decelerating, investable growth runway.

First-year FCF growth: +40%; terminal-year growth: +6%; equity discount 10.5%.

BULL • 25% PROBABILITY

$897 per share

Longer exceptional scale advantage and more persistent cash compounding.

First-year FCF growth: +55%; terminal-year growth: +10%; equity discount 9.5%.

Complete ten-year model growth assumptions

YEAR	BEAR	BASE	BULL
2027	25.0%	40.0%	55.0%
2028	23.0%	35.0%	45.0%
2029	21.0%	29.0%	35.0%
2030	18.0%	23.0%	29.0%
2031	16.0%	19.0%	23.0%
2032	14.0%	16.0%	20.0%
2033	12.0%	12.0%	18.0%
2034	10.0%	10.0%	15.0%
2035	8.0%	8.0%	12.0%
2036	6.0%	6.0%	10.0%

Scenario-weighted research value **$548**; it is not the expected October 2027 trading price. The model spans many business cycles and requires consistent evidence on pricing, demand, cash conversion and capital allocation.

08 · Portfolio fit using canonical export 63

CURRENT EXPOSURE

41 ADRs

Portfolio weight: **0.98%**. Workbook last price **$413.16** (stale versus October 19 live quote). Cost basis **$17,479.59**; average **$426.33**. Sole holdings source: export 63.

MARK-TO-MARKET SENSITIVITY

$17,229

Indicative October 19 market value using workbook quantities × independent closing price; change versus workbook value **$+289** before fees or trades. No assumption that the canonical workbook has been updated.

Portfolio role: **7 / 10** assessed separately from valuation. Adding at market prices changes portfolio concentration, correlation and retirement funding risk even if the research target suggests long-run upside. Capital-allocation authorization remains with Portfolio CIO; equity research is not a trade instruction.

09 · Catalysts, dated checkpoints & thesis-break criteria

Evidence to demand next

- Q4 revenue $52.3–54.3B; gross margin 65–67% and operating margin 56.5–58.5%.
- 2027 N2-family more-than-doubling capacity plan, yields and cash returns; monthly revenue and HPC demand.
- Incremental CoWoS/advanced-packaging capacity monetization and AI customer order diversification.
- 2027–28 U.S. fab and potential Texas commitments: firm contracts, pricing, incentives and utilization.

Explicit thesis-break conditions

- Two quarters of meaningful advanced-node capacity underutilization or falling AI/HPC demand without clear temporary causes.
- N2 yield, delivery, or qualification setbacks materially prevent anticipated node economics and customer migration.
- Structural gross margin below ~56% with sustained adverse returns from overseas construction.
- Capex remains $60B+ as sales and operating cash flows materially weaken, depressing multi-year owner FCF.
- Geopolitical conditions meaningfully interrupt Taiwan production, exports, customers or financial settlement.

> **Separation of roles:** Weakness in forward valuation alone may trigger a suggested lower purchase price or watch posture without implying the underlying company moat broke. Breaking an operating condition triggers formal CIO review; this report does not alter the official status or current position.

Outstanding diligence questions

- Are 2027/28 overseas-fab incentives effectively lowering the required investment per incremental wafer?
- What proportion of current N2 and CoWoS capacity is contractually or commercially committed?
- Will NT$/US$ exchange rates continue supporting dollar-denominated valuation assumptions?

10 · Evidence register and research limitations

Primary documents take precedence over market aggregates. S&P Global-derived StockAnalysis data supply current market capitalization, share statistics, trailing FCFE proxy and forward analyst estimates. Prices use independent October 19 regular-session closes; data snapshots around market close may differ slightly across vendors.

01 TSMC Q3 2026 earnings release / SEC

02 TSMC Q3 official results

Read source ↗

Read source ↗

03 AP report on overseas expansion
Read source ↗

04 Market valuation / statements
Read source ↗

05 S&P/StockAnalysis estimates
Read source ↗

06 Daily ADR close Oct 19
Read source ↗

Facts: company filings/management press releases, source-supported market data, user-provided holdings. **Third-party estimates:** 2027 EPS/revenue consensus and Street targets. **Analyst assumptions:** ten-year FCF growth vectors, cost of equity, terminal growth, cash normalization, case probabilities, intrinsic value and portfolio research posture.

Evidence confidence: high for filed quarter and portfolio counts; medium for consensus snapshots; low-to-medium for exact long-dated DCF outputs. **Underwriting readiness:** refreshed scenario underwrite, decision-gated by the next company-specific result. Formal research concluded; publication to PCCs not authorized. **Source-control exception:** default GitHub manifest still names export 61 despite October 16 explicit export 63 designation; no older workbook was substituted.

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